

# **Basic Network Sniffer**

## **CodeAlpha Cyber Security Internship - Task 1**

### **1. Introduction**

A network packet sniffer is a cybersecurity tool used to capture and analyze network traffic. It helps in understanding how data moves between devices and how different network protocols communicate.

In this project, a basic network sniffer was developed using Python and the Scapy library.

### **2. Objective**

The main objectives of this project are:

- To capture network packets in real time.
- To analyze packet information.
- To identify different network protocols.
- To extract useful details from network traffic.

### **3. Tools Used**

- Python 3
- Scapy
- Kali Linux

### **4. Implementation**

The program uses Scapy's sniff() function for continuous packet capturing.

For every captured packet, the program checks:

- Source IP address
- Destination IP address
- Protocol type

- Packet length
- Payload information

The program identifies TCP, UDP, and ICMP packets and displays their details.

## **5. Features Implemented**

### **Source IP Detection**

The program extracts the IP address of the device sending the packet.

### **Destination IP Detection**

The program displays the destination address where the packet is being sent.

### **Protocol Identification**

The sniffer identifies:

- TCP
- UDP
- ICMP

### **Packet Length**

The size of every captured packet is displayed in bytes.

### **Payload Extraction**

If packet data is available, the payload information is displayed.

### **Continuous Monitoring**

The program keeps capturing packets until manually stopped by the user.

## **6. Testing**

The program was tested by running:

sudo

python3

sniffer.py

Network traffic was generated using:

ping

google.com

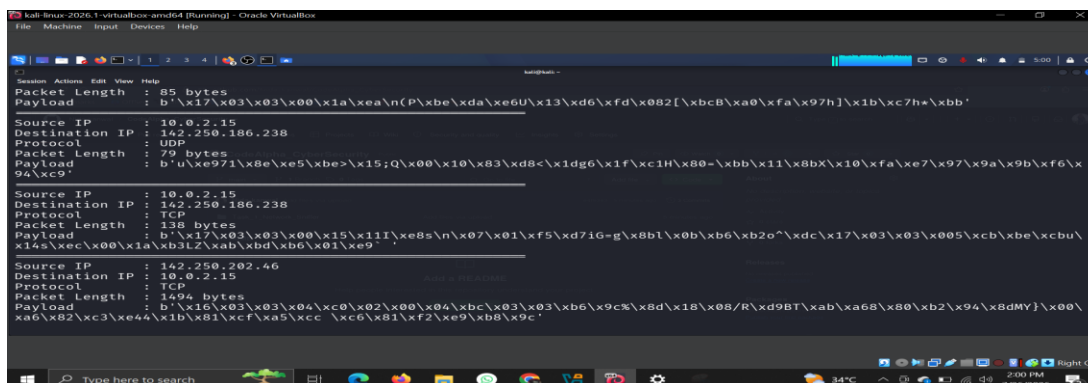
The sniffer successfully displayed packet details.

```
(kali㉿kali)-[~]
$ python --version
Python 3.13.12

(kali㉿kali)-[~]
$ sudo apt install python3-scapy -y
[sudo] password for kali:
python3-scapy is already the newest version (2.7.0+dfsg1-1).
The following package was automatically installed and is no longer required:
  libvidstab1.1
Use 'sudo apt autoremove' to remove it.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1756

(kali㉿kali)-[~]
$ nano sniffer.py
```



## 7. Result

The developed network sniffer successfully captured and analyzed live network packets. It displayed important packet information including IP addresses, protocols, packet size, and payload data.

## 8. Conclusion

This project provided practical understanding of network traffic monitoring and packet analysis. It demonstrates the basic working principles of cybersecurity tools used for network analysis and troubleshooting.

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